Navigational Vectors

Distance

1. Imagine you live in a regional town in WA and you need to plot a flight path to fly your private plane to Perth. Locate your home town on the map of WA, and draw a line between your home town and Perth airport.
2. This line represents the distance between the two towns and is a scalar value. Use the scale on the map to calculate the actual distance in km.

Length of line on map: _____ Distance in km: _____

Can you plot a flight path from your home to Perth if you know distance alone? Of course not, direction is also an important factor.

3. Using a compass rose on your map find the direction that a pilot would need to head the plane to go from your hometown to your state capital. Express your direction as a degree heading from North (0º).

_____

4. Now express the distance and direction together (example: 10 km at 37º). Distance and direction, together, is called *displacement*. Displacement is a *vector*.

_____

5. Now check your answer online.

_____

Scalars and Vectors

A scalar is just a regular number. You've been using scalars for a long time now. Your height, how many dollars you have, and how much time has gone by in this class so far -all of these are scalars. No big deal, right? The only information a scalar gets across is the size of something, which we call magnitude. Scalars only have a magnitude, or size.
A vector is like an "enhanced" number. It also has a magnitude (size), but in addition, it contains a direction, which is usually given in degrees. Because of this, a vector is written differently than a scalar. The most common way of writing a vector is to draw it using an arrow. The length of the arrow represents the magnitude, and the direction the arrow is pointing in represents...well, the direction. Here's an example:



If you took out a ruler and measured the length of the arrow that would be the vector's magnitude. And the direction is due East. (If we set North as 0º, which is what pilots do, then the vector above has a direction of 90º.)
So, now do you see why distance is a scalar and displacement is a vector? Distance is simply how far you have moved, while displacement is where you are in relation to where you started out from.

1. Draw a vector on your map from your hometown to your state capital. Is the tail or the head of the vector at the state capital? _____

2. The following information is important for pilots to know for any airplane flight. Which of these quantities are vectors and which are scalars? Why?

•Number of passengers

- Plane speed
- Plane velocity
- Distance travelled
- Flight displacement
- Flight duration
- Plane Altitude
- Acceleration of plane
- Deceleration of plane
- Amount of fuel needed

3. Give examples of other vector and scalar quantities that you use or encounter every day.

4. In what other areas other than flight would knowledge of vectors be useful?

5. Imagine that you are a pilot ready to fly between two cities. You must first file a flight plan. What do you think should go in a flight plan? Choose any two cities, devise your own version of a flight plan for this trip, and share it with the rest of the class.

Graphical Addition of Vectors

A) Your Mission: Plan a flight from your state capital to Sydney, NSW with a stopover in Coober Pedy.

1. Locate a map of the Australia and find your state capital on the map.

2. Draw a vector from your state capital to Coober Pedy. Using a ruler, compass rose or protractor, and the map, determine the flight's:

a) Distance (km)

b) Directional heading in degrees (as measured from 0º North)

c) Displacement

3. Draw another vector from Coober Pedy to Sydney. Again using a ruler, compass rose or protractor, and the map, determine this flight's:

a) Distance (km)

b) Directional heading in degrees (as measured from 0º North)

c) Displacement

4. Check your displacements for each leg of the flight online.

a) Displacement from state capital to Coober Pedy (km and degrees)

b) Displacement from Coober Pedy to Sydney (km and degrees)

B) How far has the plane travelled?
1. Find the total distance the plane travelled by adding the distance that each vector represents.

2. Does this value indicate how far the final destination is from the origin?

B) How far is the plane from your state capital at the end of the trip?

The value that you are looking for is the total displacement which is the distance and direction from the point of origin to your final destination. To determine this you must add the vectors:

Vector Addition Instructions

- Draw the first vector on your map (your state capital to Coober Pedy).
- Draw the second vector on your map (Coober Pedy to Sydney) with the tail of this vector beginning at the head of the first vector.
- Draw a third vector called the resultant from the tail of the first vector to the head of the second vector.
- Measure the length of the resultant vector (km) and its directional heading (degrees) from your state capital. This resultant represents the total displacement of the plane.

1. What is the resultant displacement of the plane from your state capital to Sydney (as measured from your vector diagram)?

2. How did your measured displacement compare with the displacement found using the internet?

1. Think of a situation (other than flying) where it would be preferable to know displacement rather than distance travelled.

2. Think of a situation (other than flying) where it would be preferable to know distance travelled rather than displacement.

3. Create a vector game or puzzle for your classmates to solve. It could be a board game, a maze, a riddle, a mystery, a treasure hunt or anything else that you wish. It should use 5 vectors. Trade games/puzzles with your classmates and see if they solve it correctly.

Wind Velocity

A) Flying in the wind

1. Imagine that you are walking in a cyclone with howling winds all around you. How will the wind affect your movement if it is blowing:

a) at your back

b) directly towards you

c) from the side

Flying a plane in the wind can produce the same effect. A pilot generally wants to fly in the shortest path between two cities. Why? Well, it saves time and fuel. The wind, however, may prevent her from doing this.

1. How can the wind affect a plane's flight?

2. What does a pilot need to do to compensate for the wind?

B)
Determining Wind Velocity
Winds in the northern hemisphere normally blow from west to east. This can change dramatically, however, when there is a front moving across the continent. Velocity is another example of a vector. Wind velocity is indicated by the wind speed and direction.
Pilots use wind maps that show the wind velocity at the altitude of the plane. A special symbol on the map called a wind barb shows the both the speed and direction of the wind (wind velocity).

C)
Resultant Velocity

1. Imagine that you are flying a plane over your home state. Your altitude is 14,000 feet. The controls in the plane are set so that you are headed due North at 400km/h. Draw a vector to represent your plane's velocity. Make sure to draw it to scale.

2. What was the wind velocity (speed in km/h and direction) over your home state at 14,000 feet?

3. Draw this wind velocity vector so that its tail is at the head of your plane's velocity vector. Make sure to draw it to scale.

4. What is the plane's resultant velocity?

5. Create a special notation or symbol to describe a river current of 4 m/s flowing south-east and explain how to read it.

6. Find examples of how wind affects sports activities, architectural designs, or other real world applications.

D)

1. A captain is heading his boat directly across a river with a very strong current. As the first mate, you must advise him how the river current will affect the boat's movement. Prepare a drawing that you can use to show the captain the boat's resultant velocity.

2. Make up a football, soccer, or other sports play that uses two or more vectors at right angles to each other. Write out the details of the play as if you were an announcer covering the game. Then give the sports play to one of your classmates and have your classmate

find the displacement of the play using the Pythagorean Theorem and one of the trig functions.